EXPERIMENT-15:

AIM:

To write a C program to perform a letter frequency attack on an additive cipher and display possible plaintexts in order of likelihood, allowing the user to specify the top \(N\) results.

CODE:

```
#include <stdio.h>

#include <ctype.h>

#include <string.h>


int main() {

    char c[1000];

    int n, k, i;

    char *freq = "ETAOINSHRDLCUMWFGYPBVKJXQZ";


    printf("Enter ciphertext: ");

    fgets(c, sizeof(c), stdin);

    printf("Enter top N plaintexts: ");

    scanf("%d", &n);


    for(k = 0; k < 26 && k < n; k++) {

        printf("\nShift %d: ", k);

        for(i = 0; c[i]; i++) {

            if(isalpha(c[i]))

                putchar((tolower(c[i]) - 'a' - k + 26) % 26 + 'a');

            else

                putchar(c[i]);

        }
```

```
    }
    return 0;
}
```

OUTPUT:

